

MARCH, 2025

TECHNICAL REGULATIONS

2026

COMPETITION
RULES



MAJOR CHANGES

GET READY FOR

THE MOST EXTREME SOLAR CHALLENGE!

4M² AND 6M² ARRAY SIZES,
SOLAR COLLECTOR SIZE,
ENERGY ALLOWANCE, 3
WHEELS, 2 PASSENGERS,
GROUND CLEARANCE,
HADLAMPS

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**SOLAR
CHALLENGE**

PUBLISHED BY SASOL SOLAR CHALLENGE



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Introduction and Welcome

Dear Solar Champions,

It is our privilege to welcome you to the 2026 Sasol Solar Challenge—an event that not only tests the limits of solar mobility but also celebrates the spirit of human ingenuity and endurance.

Africa, the ancient continent, has long been a land of discovery, resilience, and boundless potential. As we welcome teams from across the world, we invite you back to the Cradle of Mankind, where innovation meets heritage, and where the future of sustainable mobility takes centre stage.

Our goal as organisers is simple yet profound—to create an experience that is life-changing for all who take part. Your dedication, hard work, and relentless pursuit of excellence inspire us, and we are committed to delivering a challenge worthy of your time, effort, and vision. The impact of your work extends far beyond the competition itself. As your solar cars navigate through the diverse landscapes of South Africa, they ignite curiosity, inspire future engineers, and leave a lasting impression on the communities we visit.

For 2026, we continue to push the boundaries with our Challenger, Cruiser, and Innovation classes—each providing a platform for groundbreaking technology, strategic brilliance, and pioneering spirit. Whether harnessing solar energy or advancing the future of New Energy Vehicles, you are shaping the evolution of sustainable transport, and we are honoured to witness your journey.

This challenge is a crucible of innovation—an arena that invites all who dare to push the limits of what is possible. It will test you in every way, demanding endurance, resilience, and ingenuity. But it will also reward you with moments of triumph, discovery, and the unbreakable camaraderie that comes from facing the ultimate test together.

As organisers, we share your passion for the pursuit of excellence. We cannot wait to chase the horizon with you—fuelled by the energy of the sun and driven by the unrelenting spirit of innovation.

Welcome to the world's most extreme solar car challenge!

Robert Walker
Sasol Solar Challenge Event Director





The major changes for 2026 include:

- Increased solar collector sizes and reduced energy storage allowances for 6m² Challenger and Cruiser vehicles
- Challenger Class Vehicles will include 4m² (based on 2024 Sasol Solar Challenge Regulations) and 6m² (based on 2025 Bridgestone World Solar Challenge Regulations) array size Competition Vehicles
- Cruiser vehicles may have three wheels
- All Competitions Vehicles must be fitted with headlamps
- Energy storage measured by energy rather than mass for 4m² and 6m² Challenger and Cruiser Class Vehicles
- Easier-to-follow Cruiser format—fixed energy allowance, two occupants, smart design features
- Front and rear overhang restrictions, and new ground clearance requirements.
- Official Start and Official Finish days moved from Friday-schedule to Thursday-schedule

The information contained in this document is for the purposes of conducting the 2026 Sasol Solar Challenge and must not be regarded as constituting definitive instructions as to how a Competition Vehicle should be constructed or operated.

Additional information or clarification of the intent of regulations is included in *italics*.

Please pay meticulous attention to all elements of this document to ensure your team's compliance with these regulations.

Special Credits

Technical Regulations ©Bridgestone World Solar Challenge, used with permission.

1. Competition Vehicle Regulations

1.1. Classes

- 1.1.1. The 2024 Sasol Solar Challenge will have four classes of Competition Vehicle:
- The two Challenger Classes are for single-seat Competition Vehicles optimised for converting solar irradiance to speed (4m² and 6m² array sizes)
 - Cruiser Class is for Competition Vehicles with two or more seats that will be judged on practical design features as well as efficiency.





- Innovation Class is open to all vehicles using renewable energy as their primary source, including prototypes and OEMs of NEVs.

4m² Challenger Class regulations are based on the 2024 Sasol Solar Challenge Regulations, 6m² Challenger Class regulations are based on the 2025 Bridgestone World Solar Challenge Regulations.

*It is accepted that energy from regenerative braking, road gradients and wind incident on the Competition Vehicle while driving may also be used. Active aerodynamic features are permitted but must not reduce the stability of the vehicle. **Cruiser** Competition Vehicles must be designed to be powered by renewable energy but may be recharged from non-renewable sources if renewable sources are not available.*

1.2. Dimensions

- 1.2.1. When driving in a straight line, the Competition Vehicle must fit inside a right rectangular prism 5800 mm long, 2300 mm wide and 1650 mm high, with the tyre contact points on the base of the prism.

Standard 20' shipping containers have 5880 mm internal length and 2350 mm internal width.

1.3. Wheels

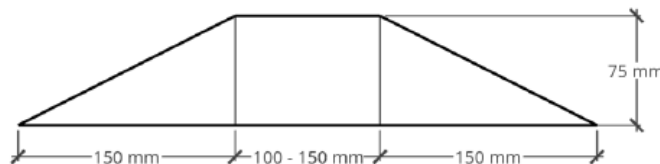
- 1.3.1. Challenger Class and Cruiser Class Competition Vehicles must be supported by at least three wheels while driving.
- 1.3.2. The longitudinal distance between the frontmost part of the Competition Vehicle and the frontmost tyre contact point must be less than 60% of the wheelbase.
- 1.3.3. The longitudinal distance between the rearmost tyre contact point and the rearmost part of the Competition Vehicle must be less than 60% of the wheelbase.
- 1.3.4. The fully laden Competition Vehicle must be able to drive over a road apex formed by a 1:15 incline followed by a 1:15 decline without any part of the Competition Vehicle other than the tyres touching the road.



The length of the road apex will be greater than the wheelbase of the Competition Vehicle.



- 1.3.5. The fully laden Competition Vehicle must be able to drive over a speed hump with the dimensions shown in the following diagram without any part of the vehicle, other than the tyres, touching the speed hump.



- 1.3.6. The fully laden Competition Vehicle must have an approach angle of at least 10° and a departure angle of at least 10° .

The Competition Vehicle must comply with all regulations in Section 1.3 whenever it is being driven.

1.4. Solar Collector

- 1.4.1. Competition Vehicles may incorporate a solar collector that converts solar irradiation into energy that can be used to propel the vehicle. The most common type of solar collector for Competition Vehicles uses photovoltaic cells. Teams wishing to use other types of solar collector must send details of their proposed solar collector to the Event Organiser for approval.
- 1.4.2. A 4m² Challenger vehicle must not have more than 4.000 m² of solar collector area.

Example calculation: The area of a SunPower cell with a width of 125 mm and a diagonal diameter of 166 mm is less than 0.0155063 m², and so the area of 257 cells is less than 3.9852 m².

- 1.4.3. A 6m² Challenger vehicle must not have more than 6.000 m² of solar collector area.

Example area calculation: The area of a SunPower cell with a width of 125 mm and a diagonal diameter of 166 mm is less than 0.0155063 m², and so 386 cells will have a total area less than 6.000 m².

- 1.4.4. Solar collector area includes:
- the directly exposed front surface area of all photovoltaic cells
 - the directly exposed rear surface area of all bifacial photovoltaic cells
 - the directly exposed surface area of energy conversion devices other than photovoltaic cells
 - the directly exposed area of all surfaces that transmit, refract or reflect light onto devices that convert solar irradiation into energy that can be used to propel the vehicle.

A surface is directly exposed if light can travel in a straight line from any part of the sky, ground or road to the surface.





PV cell encapsulation within 10 mm of a minimum rectangle covering a PV cell will not be counted towards the solar collector area.

The solar collector area limit applies to all possible configurations of the solar collector, while driving and while solar charging.

Photovoltaic cell area includes active material, busbars, fingers, connection pads, and cell area covered by interconnection ribbon or tabs.

If a bifacial photovoltaic array is tilted for solar charging so that the front and rear surfaces are both directly exposed, then the front and rear surface areas are both counted.

It is accepted that light reflection from parts of the Competition Vehicle such as the canopy is unavoidable.

- 1.4.5. The solar collector area for Cruiser vehicles is unrestricted, but the solar collector must be within the allowable dimensions of the Competition Vehicle.
- 1.4.6. Materials with high environmental toxicity, including GaAs, CdTe and CuInSe₂, must not be used for solar energy conversion.
- 1.4.7. All devices used for solar charging must be carried in the Competition Vehicle. This includes stands, supports, and cables.
- 1.4.8. During solar charging, Competition Vehicles must not be tilted on jacks or stands that were not carried in the vehicle even if you are pretending to change a wheel.

1.5. Energy Storage

- 1.5.1. A Competition Vehicle may store energy. A standard energy storage system uses rechargeable electrochemical cells. Teams wanting to use some other form of energy storage must send details of the proposed energy storage system to the event organisers for approval.
- 1.5.2. Teams wanting to use other cell chemistries, or a mix of cell chemistries must send details of their proposed energy storage system to the event organisers for approval. The Chief Energy Scientist will determine the allowable configurations.
- 1.5.3. Energy storage capacity must be no more than:
 - 15MJ for 4m² Challenger vehicles
 - 11MJ for 6m² Challenger vehicles
 - 55MJ for Cruiser Vehicles
- 1.5.4. The energy storage capacity of electrochemical cells will be based on specifications endorsed by the cell manufacturer. The nominal energy of a cell, in joules, will be calculated by E





$$E = 3600 V.q$$

where V is the nominal discharge voltage of the cell, in volts, specified by the manufacturer, and q is the maximum of the:

- rated discharge capacity
- minimum discharge capacity
- typical or nominal discharge capacity, in Ah, specified by the manufacturer.

- 1.5.5. The energy storage capacity for other forms of energy storage will be determined by the Chief Energy Scientist and will be based on energy that can be delivered from the 'tank'.

The remaining regulations in this section are based on electrochemical energy storage. The Event Organiser will require that alternative energy storage systems meet equivalent levels of safety based on the organiser's assessment of the proposal submitted by the Team.

- 1.5.6. Specifications from third party suppliers found on the internet might not match those endorsed by manufacturers. If the Event Organiser receives conflicting or unclear specifications of electrochemical cell energy for a cell model, the Chief Energy Scientist will determine the nominal cell energy for cells of that model.
- 1.5.7. The energy storage system must be contained within at most two packs. Energy storage packs must be rigid, self-contained boxes that cannot be opened or disassembled without using tools and can be removed from the Competition Vehicle intact.
- 1.5.8. Electrochemical cells must not, at any time, be operated outside of the operating ranges for voltage, current, and temperature specified by the manufacturer. Teams must provide the manufacturer's specifications that include:
- Minimum operating cell voltage
 - Maximum operating cell voltage
 - Maximum discharge current
 - Maximum charge current
 - Maximum temperature while discharging
 - Minimum temperature while charging
 - Maximum temperature while charging
- 1.5.9. The Competition Vehicle must automatically prevent electrochemical cells from being operated outside the operating ranges for voltage, current, and





temperature specified by the manufacturer. Teams must provide an endorsement by their certifying engineer that an adequate and effective automatic battery management system has been designed and implemented so that fault conditions will be managed safely.

1.5.10. Batteries used only to

- Power a real-time clock when the Competition Vehicle is turned off; or
- Retain data when the Competition Vehicle is turned off; or
- Power wireless tyre pressure monitors

are not considered to be part of the energy storage system, provided that the total energy capacity does not exceed 10 kJ.

1.5.11. Batteries or cells inside devices such as handheld radios, cameras, mobile telephones, or wristwatches that are carried by the driver or passengers are not considered to be part of the energy storage system if they are not electrically connected to the Competition Vehicle, its instrumentation or control systems.

1.5.12. If a commercial-off-the-shelf phone or tablet containing a battery is used as a driver display, it must be powered from the Competition Vehicle's electrical system. The electrical connection must be done in such a way that power cannot flow from the device back into the Competition Vehicle. The blocking diode or other system used to achieve this must be mounted within 200 mm of the device. The device must not be charged from an external power source during the Event. The device's battery capacity will be counted as part of the energy storage capacity. The device will not be considered to be an energy storage pack.

It is recognised that failure of the blocking device could result in 5 V being present throughout the Competition Vehicle even in safe state. This is acceptable as long as:

a) the team can demonstrate that such a failure is highly improbable and

b) that in the event of such a failure, no combination of additional failures can prevent the Competition Vehicle from entering safe state.

1.5.13. Capacitors are not considered to be part of the energy storage system if their total energy capacity is less than 36 kJ. Such capacitors must be automatically discharged to less than 60 V within five seconds of the Competition Vehicle being placed in safe state (see Technical Regulations 1.29).

1.5.14. Energy storage packs must be mounted in the Competition Vehicle so that they will be restrained in a 20 g acceleration in any direction.





- 1.5.15. The Competition Vehicle must be designed so that any liquids or gases from a damaged energy storage pack cannot enter the occupant space.
- 1.5.16. Energy storage packs must be constructed so that each pack can be sealed using tamper-evident plastic seals, like 3 × 100 mm plastic cable ties. With seals fitted, it must not be possible to remove any cell from a pack without breaking the seal. Seals will be provided by and fitted by the event organiser at scrutineering.
- 1.5.17. Energy storage packs of Challenger vehicles must have a means of being sealed so that no electrical connection can be made to the energy storage system. The seals will be tamper-evident plastic seals, like 3 × 100 mm plastic cable ties. Seals will be provided by Officials and/or observers.

This regulation allows teams to have the energy storage system impounded inside the Competition Vehicle. Examples of acceptable sealing mechanisms include:

- *Disconnecting each energy storage pack from the rest of the Competition Vehicle and sealing a cover over each pack that prevents access to all electrical connectors and to the inside of the pack*
 - *Removing the energy storage packs from the Competition Vehicle and sealing them in a box provided by the team*
- 1.5.18. Energy storage packs must be in “safe state” while being removed from the Competition Vehicle and while being reinstalled into the Competition Vehicle.
- 1.5.19. Energy storage packs must be designed and constructed so that scrutineers can verify the cell models being used and the number of cells of each model.
- 1.5.20. Any external charging system that is used to recharge the energy storage system must meet the following requirements:
- The charger must be used with a residual current device
 - The charger must be either permanently connected to the energy storage system, or connect to the energy storage system using an appropriate connector
 - The output of the charger must be electrically isolated from any AC input
 - Charging must stop automatically when the energy storage system is full or if a fault occurs

This regulation applies to all charging done in South Africa, including in the pits at Red Star Raceway.

1.6. Cruiser Charging

- 1.6.1. Cruiser vehicles may be recharged from an external energy source overnight, during the time intervals from the time Parc Fermé opens after the day’s finish until 06H00 the following morning when the energy storage packs will be





unsealed by the Officials or observers, each day. (Sporting Regulation 2.30.9., 2.30.10. and 3.2.9.)

- 1.6.2. Each Cruiser Team is responsible for providing the equipment required to recharge their Competition Vehicle.
- 1.6.3. Cruiser Class Competition Vehicles must be equipped with an on-board AC charger with an IEC 62196-2 Type 2 (male) charging inlet and be capable of charging from a single-phase AC supply (230 Vac, +10%, -6%, 50 Hz). The maximum current drawn from the supply shall be 15A, although the organisers retain the right to reduce this maximum current limit at any time for operational reasons.
- 1.6.4. Battery swapping (or the equivalent for other types of energy storage) is not permitted.

1.7. Vehicle Identification

- 1.7.1. The Competition Vehicle must have a unique identifier, which must be permanently attached to a substantial part of the Competition Vehicle chassis or frame. The identifier must include a manufacturer code, the year of manufacture, and a serial number.

Teams may use a Vehicle Identification Number (VIN) or generate their own identifier, e.g., SSC-2026-01.

- 1.7.2. The Competition Vehicle must have a space at least 220 mm wide and at least 110 mm high for mounting a rear vehicle registration plate. Whenever the Competition Vehicle is being driven:
 - the registration plate must be mounted in a vertical plane with its longest edge
 - horizontal and perpendicular to the direction of travel
 - the letters must be upright
 - the registration plate must be within 50 mm of the rearmost part of the Competition Vehicle
 - no part of the Competition Vehicle may be in the region rearward of the registration plate in the range 15° up, 0° down, 45° left and 45° right from all parts of the registration plate.
- 1.7.3. You must not modify the registration plate.

You must not enlarge the holes in the registration plate. You must not use the existing holes for rear vision.

1.8. Signage





- 1.8.1. Competition Vehicles must incorporate event signage in unbroken rectangular spaces 200 mm high and 500 mm wide on both the left and the right sides of the Competition Vehicle. Artwork will be available from the Sasol Solar Challenge website.

'Unbroken' means that the signage space must not overlap panel gaps or visible joints.

Teams must provide and install their own signage. Incorrect use of artwork will result in a sticker of the correct size and with the correct artwork being applied.

- 1.8.2. Teams must display their team number on the left and the right sides of the Competition Vehicle, in digits that are more than 150 mm high and that are clearly visible against their background.
- 1.8.3. Event signage and team numbers must be completely visible from 3 m perpendicular to the side of the Competition Vehicle and at a viewing height of 1.8 m above the ground.
- 1.8.4. Competition Vehicles must have a front signage area on the Competition Vehicle body, forward of the windscreen. The front signage area must contain:
- The name of the Competition Vehicle, or the name of the team, or the number of the Competition Vehicle, readable at 3 m perpendicular to the centre of the name
 - A 150 × 150 mm square event logo

The entire front signage area must be visible in top view, in front elevation view and must not overlap with the solar collector. Artwork for the event logo will be available on the Sasol Solar Challenge website, or a sticker for the event logo will be provided by the event organisers only upon request and approval.

- 1.8.5. The national flag of the country of entry must be displayed on the car, adjacent to the windscreen. The minimum size is 70 mm × 40 mm. The flag must not be broken.

1.9. Ballast

- 1.9.1. Each Challenger and Cruiser Class Competition Vehicle occupant will be assigned ballast so that the combined mass of the occupant and their ballast is at least 80 kg.

Innovation Class team occupants need not carry ballast and will not be weighed.

- 1.9.2. Ballast must be supplied by the team and must be fashioned in such a manner to ensure the weight is tamper-proof and made from a durable material to eliminate the chances of breakage or damage.





All ballast will be weighed during scrutineering to verify the weight. If a ballast weight discrepancy is found during scrutineering, the event organisers will provide the team with supplementary ballast to make up the correct weight. Ballast may not consist of any functioning item of the Competition Vehicle such as batteries etc. and must be non-functional.

- 1.9.3. Teams must ensure each ballast weight is clearly indicated on the surface of the ballast. The weight indication must not wear down or disappear when being handled roughly.

Each team must supply the appropriate amount of ballast for each individual Competition Vehicle occupant, sharing of ballast between occupants is not permitted when occupants are driving in the Competition Vehicle during competition hours.

- 1.9.4. Challenger and Cruiser Class Competition Vehicles must have means of securing ballast within 300 mm of each occupant's hip point.
- 1.9.5. Supplementary ballast bags provided by the event organisers must be returned to the pack down area at the end of the event.

1.10. Tracker

- 1.10.1. The Competition Vehicle must have provision to carry a self-contained tracker provided by the event organisers.

The tracker box will be 120 mm long, 120 mm wide and 70 mm high. The tracker will be self-powered. It will not require any electrical connection from the Competition Vehicle. The mass of the tracker will not exceed 2 kg. It will emit radiofrequency energy with no more power than emitted by a mobile phone. The unit will be installed by the technical specialist team during scrutineering.

- 1.10.2. When the tracker box is installed in the car, it must be possible to construct a right circular cone with its axis perpendicular to the top face of the box and an apex angle of 120° , and with the cone touching each edge of the top face of the box, so that no ray from the apex and within the cone passes through the ground or through any part of the car that is not radio transparent at frequencies between 300 and 3000 MHz. The tracker box may be tilted within the Competition Vehicle.

Fibreglass and aramid composites, vinyl stickers, paint, clear acrylic, and clear polycarbonate are radio transparent if they do not include metal or carbon. Solar cells, carbon fibre composites and metal are not considered radio transparent.





- 1.10.3. The tracker unit must be returned to the event organisers at the end of the event.
- 1.10.4. Cruiser teams must upload their telemetry data at each daily finish line and in Cape Town at the official finish line.
- 1.10.5. The telemetry data must include the following measurements taken at 10-second intervals while driving and while charging:
- Distance travelled
 - Cumulative energy generated from the solar collector
 - Cumulative energy from the battery (increases when energy is taken from the battery, decreases when the battery is charged)
 - CAT time or GMT +2 Time

Data forms can be created in any manner as teams see fit as long as the data can be easily interpreted by the Clerk of the Course and the Chief Scientist.

Take note that telemetry data needs to be analysed and interpreted by the event organisers within 5 minutes or less. Complex data tables and graphs are not recommended.

- 1.10.6. Tracker and telemetry data will be published by the event organisers.

1.11. Safety

- 1.11.1. Teams are responsible for the safety and roadworthiness of their Competition Vehicles. Compliance with the regulations and passing scrutineering does not mean that a Competition Vehicle is safe, roadworthy, and fit for purpose.
- 1.11.2. Each team must engage a professionally qualified engineer to certify that the Competition Vehicle is designed and constructed using sound engineering practice, meets the design parameters where stated, maintains stability while driving during testing, and is roadworthy and fit for being driven from Sasolburg to Cape Town on public roads. The certifying engineer must be qualified to certify vehicles for operation on public roads.

The Roadworthiness Certificate must be signed by a single certifying engineer. However, that engineer may engage other engineers to advise on aspects outside their expertise.

A certifying engineer must have a four-year bachelor's degree or higher in mechanical or automotive engineering, five years of postgraduate experience working in the automotive industry, and professional indemnity and public liability insurance. Other jurisdictions will have similar requirements.

- 1.11.3. Any changes made after certification to items described in the roadworthiness certificate must be re-approved by the team's certifying engineer.
- 1.11.4. All parts of the Competition Vehicle must be fixed so that they cannot detach while driving. Latched parts such as doors, canopies, and top shells must be



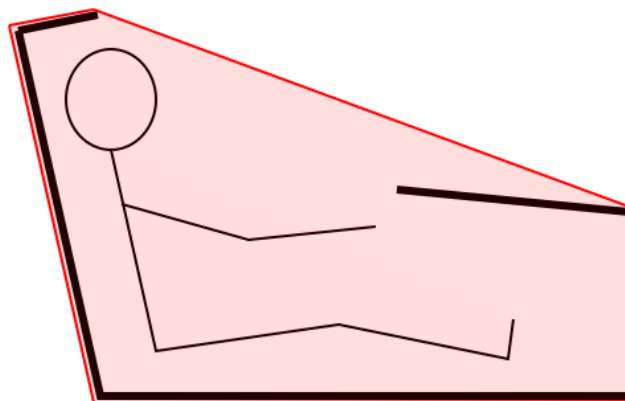


secured with two-stage latches where the first stage engages easily and is sufficient to hold the part if the second stage is not engaged properly.

1.12. Occupant Cell

- 1.12.1. Competition Vehicle occupants must be enclosed in an occupant cell designed to protect them from injury.
- 1.12.2. Vehicle components within the occupant cell must be designed to minimise the risk of injury or entrapment in a crash.
- 1.12.3. Teams must provide documentation that specifies which parts of their Competition Vehicle constitute the occupant cell.
- 1.12.4. When occupants are seated normally, with safety-belts and helmets on, they must be completely inside the convex hull of the occupant cell.
- 1.12.5. There must be a gap of at least 50 mm between every point on every occupant's helmet and the convex hull of the occupant cell.

Imagine stretching a rubber skin around the occupant cell; no part of any occupant may touch the skin, and helmets must be more than 50 mm from the skin.



- 1.12.6. Each team must provide a description of how the occupant cell will protect the occupants from frontal impacts, side impacts and rollover impacts. This description must be endorsed by the team's certifying engineer. For teams wishing to do finite element analysis of the occupant cell, the minimum test loads are:
 - Frontal impact: a 5 g load, opposing the direction of travel, applied to the front of the occupant cell in an area less than 250 mm high and less than 600 mm wide



- Side impact: a 5 g load into the side of the occupant cell, applied adjacent to the driver's torso in an area less than 250 mm high and less than 600 mm wide
- Top impact: a load with component 5 g down, 1.5 g sideways and 4 g backwards, applied at each possible area of contact between the occupant cell and the ground when the occupant cell is upside down; the contact area for each test load must have a diameter less than 150 mm

1.12.7. The occupant cell must be designed to protect each occupant's head, arms and upper torso by deflecting objects such as body panels and fencing wires away from the occupant. The structure used to achieve this must extend to the top of the occupant cell and be capable of resisting or deflecting loads of 5 g applied to it from any horizontal direction by a horizontal object 2.3 m wide and less than 250 mm high.

It is permissible for the windscreen to break provided the occupants are protected.

Technical Regulation 1.17.2 defines the maximum obscuration of the driver's vision by A-pillars.

1.12.8. Loads in Regulations 1.12.6 and 1.12.7 are based on the fully laden mass of the

1.12.9. Competition Vehicle. Teams must be able to show that the occupant cell structure will not fail with these test loads, that any deformations as a result of these loads will not impinge on the occupants, and that risks of impacts of the occupants with the interior of the Competition Vehicle are minimised.

1.13. Seats

1.13.1. Challenger vehicles must have one seat. Cruiser vehicles must have at least two seats.

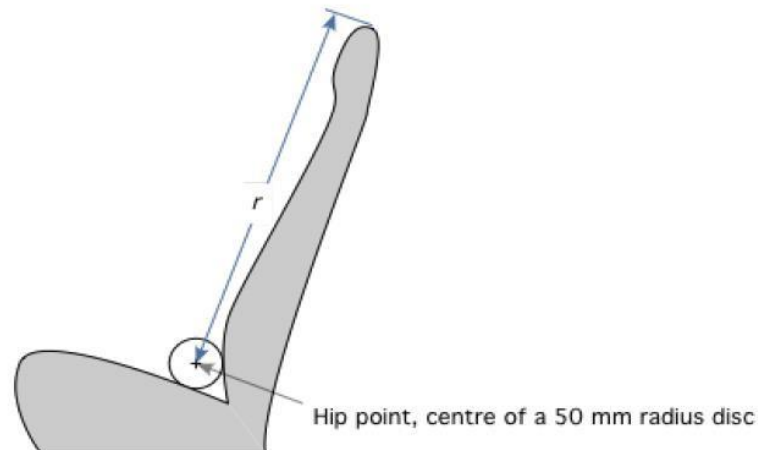
Cruiser vehicles must have two occupants while driving.

1.13.2. Each Competition Vehicle occupant must have a seat that faces forwards. The driver's legs, viewed from above, must be within 10° of the forward direction.





- 1.13.3. Each seat must have a back and a head restraint that is behind the occupant's head. The distance from the hip point to the top of the head restraint must be at least 800 mm for front seats and at least 750 mm for rear seats (UNECE Regulations 17 and 25). The hip point may be approximated as shown in the diagram below.



You may use different cushions for different drivers. The full set of cushions will be considered part of the Competition Vehicle and must be carried in the Competition Vehicle.

- 1.13.4. An Event manikin ("PVC Pat") must fit into the driver's seat and any passenger seat that will be occupied during the Event. Dimensioned drawings of the manikin will be made available on the Event website.

PVC Pat's dimensions are based on 95th percentile international male and female dimensions from the [DINED anthropometric database](#)

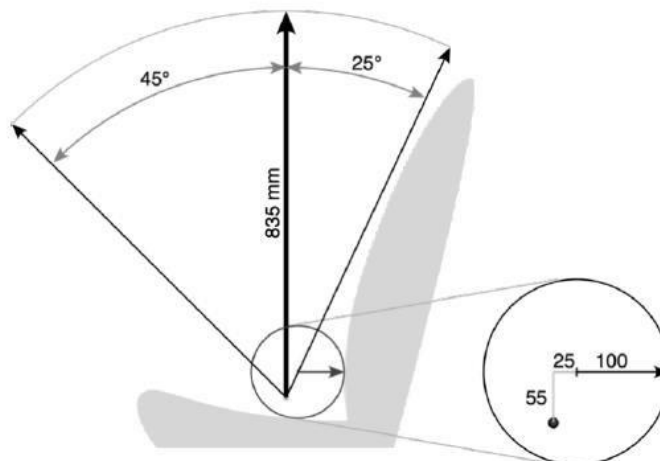
- 1.13.5. When seated in the normal seating position, PVC Pat must be protected by the occupant cell (Regulation 1.11), with the exception that PVC Pat does not have to wear a helmet (Regulation 1.11.4).
- 1.13.6. PVC Pat does not have to meet the vision requirements.
- 1.13.7. Each occupant's heels, including PVC Pat's heels, must be below their hip point.
- 1.13.8. The outer edges of PVC Pat's ankles will be 436 mm apart, as shown in the drawing. The outer edges of PVC Pat's wrists must be able to be placed 496 mm apart.
- 1.13.9. The angle between PVC Pat's shoulder, hip and knees must be more than 90°.
- 1.13.10. PVC Pat's head can tilt forwards, but no more than 30° from the line of the spine. PVC Pat's head does not yaw.
- 1.13.11. PVC Pat, or a person no smaller than Pat, must be able to get in and out of the normal seating positions. You may remove Pat's legs, temporarily, to help get the torso and head into position.





- 1.13.12. PVC Pat's hip point must be in the same position as the hips of an actual driver in the normal driving position, or an actual passenger in the normal seated position.
- 1.13.13. Head space for each seat must comply with Act 93 of 1996: National Road Traffic Act, as shown in the following diagram. The 835 mm radius arm must be able to move 45° forwards, 25° backwards, and 7° on either side of vertical. The steering wheel, mirrors, seat backs, and head restraints may be inside the occupant space but must be designed to minimise the risk of injury in a crash. No other part of the Competition Vehicle structure, including the windscreen, may be inside the occupant space.

This minimum occupant space requirement is based on a 50-percentile male and does not allow for a helmet. Taller team members may need more space.



1.14. Safety Belts

- 1.14.1. Safety-belts must be fitted for each seating position. Safety-belts must be compliant with at least one of the following standards: UNECE Regulation 16, US FMVSS 571.209, SFI 16.1, SFI 16.5, SFI 16.6, FIA 8853-2016, FIA 8854/98.

The occupant cell will provide the greatest protection when occupants are secured into the cell with four-point or five-point harnesses.

- 1.14.2. Safety-belts must be fitted and used according to the manufacturer's instructions, or as approved by the team's certifying engineer.
- 1.14.3. Safety-belt anchorages must meet the intent of UNECE Regulation 14. In particular:



- Upper anchorages for each seat must withstand a force of 13.5 kN applied to the upper safety-belt straps
- Lower anchorages for each seat must withstand a force of 13.5 kN applied to the lower safety-belt straps
- The location of anchor points must comply with the instructions of the safety-belt manufacturer, or with UNECE Regulation 14 Annex 3.

1.14.4. Compliance must be confirmed by the team's certifying engineer.

1.15. Egress

- 1.15.1. Teams must demonstrate that all occupants can exit the Competition Vehicle in less than 15 seconds, without assistance.
- 1.15.2. Doors and canopies used for egress must be capable of being secured and released from inside the Competition Vehicle and from outside the Competition Vehicle.
- 1.15.3. Emergency openings, and the methods of opening, must be clearly indicated on the exterior of the Competition Vehicle, and be visible to an emergency service first responder.

Instructions may include diagrams.

- 1.15.4. Occupants must be able to exit the Competition Vehicle without assistance.

Teams may not use adhesive tape to secure exits.

1.16. Ventilation, Cooling and Hydration

- 1.16.1. The Competition Vehicle must have sufficient ventilation to prevent the build-up of CO₂ in the cabin. Occupants must have ventilation or cooling sufficient to ensure that they will not overheat. The Team must describe the ventilation and cooling systems and have them approved by their certifying engineer.
- 1.16.2. Each Competition Vehicle occupant must have space for at least two litres of drinking water (see Sporting Regulation 2.23.3).

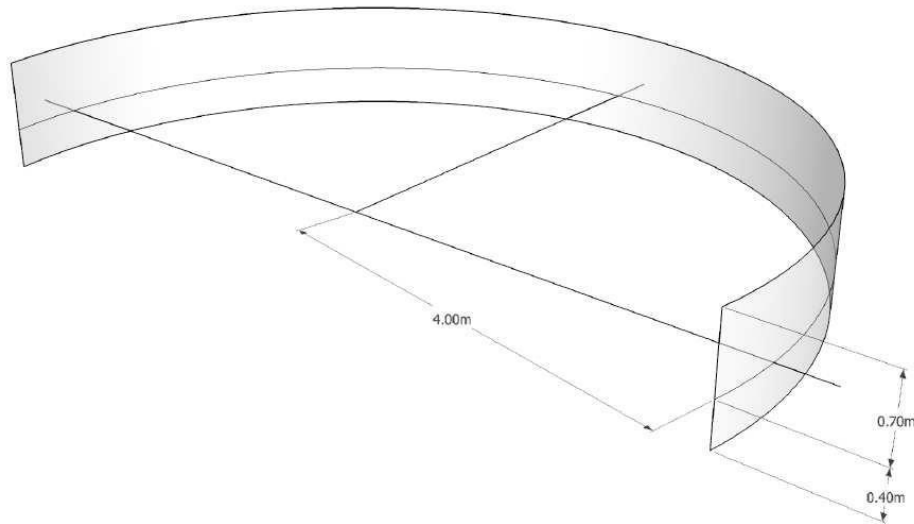
1.17. Forward and Sideward Vision

- 1.17.1. Each driver, when seated in the normal driving position with safety-belt and helmet on, must be able to identify 75 mm high letters at every point of forward travel that is:
- 4m from the driver's eyes, and
 - between 0.4m below eye level and 0.7m above eye level, and





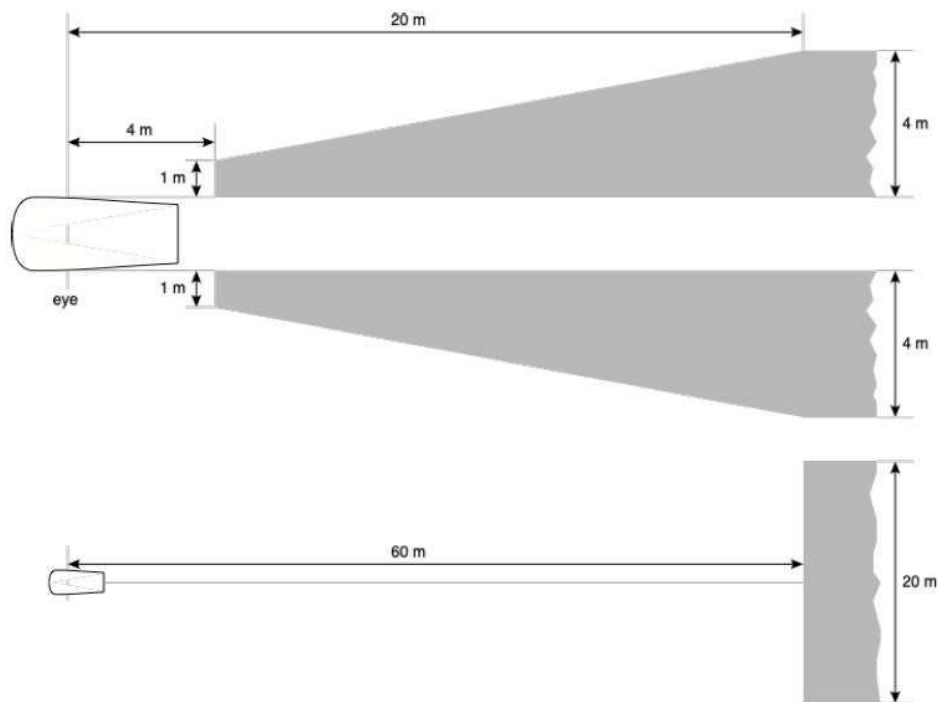
- between 100° left and 100° right of the direction of travel



- 1.17.2. There must be no more than 2 pillars within the forward and sideward field of view. Each pillar must obscure less than 6° of the horizontal field of view.
- 1.17.3. Forward and sideward vision must be achieved without the aid of mirrors, lenses, or electronic vision systems.
- 1.17.4. The windscreen that is used for forward and sideways vision must have an optical transmittance of more than 75%.
- 1.17.5. Traffic light colours must be discernible through the windscreen.

1.18. Rear Vision

- 1.18.1. The Competition Vehicle must have rear vision systems that enable the driver, when seated in the normal driving position with the safety belt fastened, to see the ground in the shaded areas shown in the diagrams below (UNECE Regulation 46, Section 15).



- 1.18.2. Rear vision systems may be electronic, mirrors, or both. Rear vision systems must operate whenever the Competition Vehicle is in motion under its own power or about to be driven. Rear vision images must be oriented so that objects on the right of the Competition Vehicle are on the right of the image.

1.19. Steering

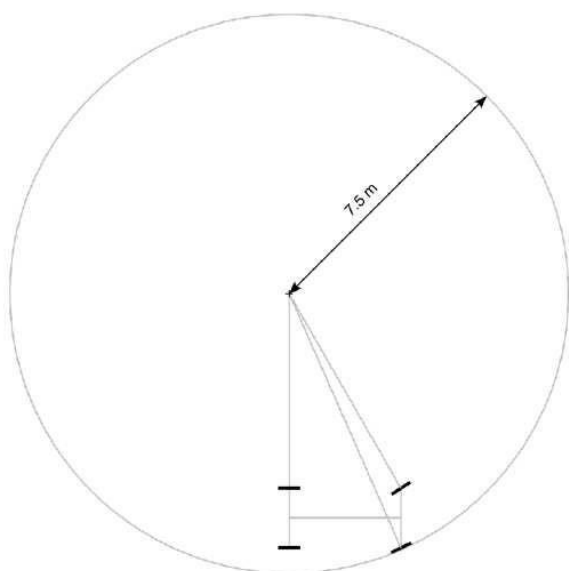
- 1.19.1. Steering must be controlled by a steering wheel designed so that it cannot catch on any clothing while driving or when the driver exits the Competition Vehicle.
- 1.19.2. Failure of any non-mechanical component of the steering system must not prevent effective steering of the Competition Vehicle.
- 1.19.3. Steering shafts must be designed to reduce the risk of injury to the driver in a crash.

A collapsible boss is an acceptable method to reduce steering wheel impacts.

- 1.19.4. Rear-wheel steering is not permitted. Specifically, no wheel that is rearward of the centre of mass of the Competition Vehicle may be steered, either by a mechanism that the driver operates or via a servo motor. This does not preclude suspension designs that are deliberately designed to have built-in bump steer or roll steer.



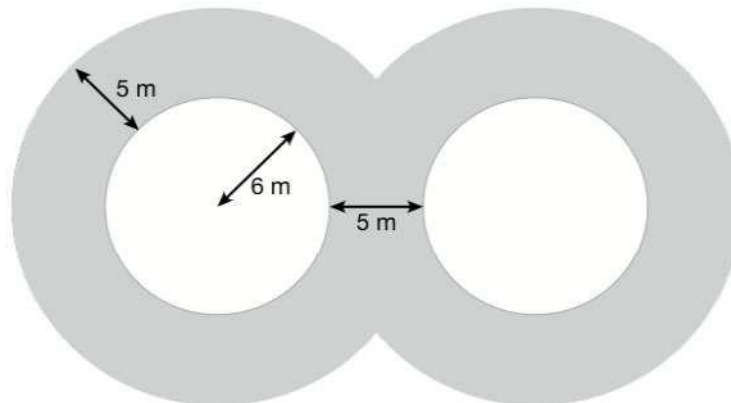
- 1.19.5. The Competition Vehicle must be able to turn in each direction with a circle with a radius of 7.5 metres, measured to the extreme outer edge of the tyre track at ground level.



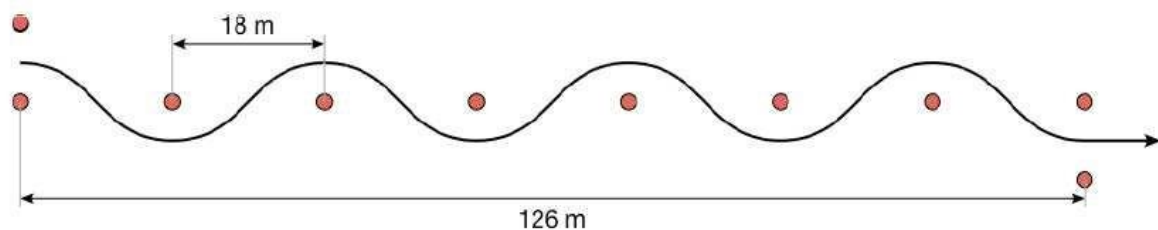


1.20. Stability

- 1.20.1. Competition Vehicles must be able to negotiate a Figure-8 course in less than 9 seconds per side and less than 18 seconds overall.



- 1.20.2. Competition Vehicles must be able to negotiate a slalom course in less than 11.5 seconds.



- 1.20.3. The Competition Vehicle must be stable at all achievable speeds and in crosswinds likely to be encountered. Teams must provide analysis or test results that show that the car will be stable at high speeds and in crosswinds. This analysis must be approved by the team's certifying engineer.

In recent solar car events, at least six cars left the road when hit by strong wind gusts. Wind gusts exceeding 100km/h are possible. Some of the factors affecting vehicle stability are presented in the informal discussion paper by Prof John Storey, "Stability Considerations", which will be available on the Sasol Solar Challenge website.

- 1.20.4. The fully laden Competition Vehicle should not tip when tilted by 45° about each pair of adjacent tyre contact patches, in all driving configurations.

The Team's certifying engineer must approve the stability and the rollover propensity of the Competition Vehicle. A tilt test is one way of testing the static stability factor (SSF) of a Competition Vehicle, which is one indication of susceptibility to rollover. Notice that the Regulation says 'should' and not 'must'. It is permissible to have a Competition Vehicle that does not pass a static tilt test, provided that the Competition Vehicle has some other mechanism for handling the equivalent forces without tipping. If the Competition Vehicle cannot pass a tilt test in all suspension configurations it becomes particularly important that the certifying engineer





be satisfied with the roll-over stability under all conditions. Their certification documentation should provide a summary of analysis and testing you have done.

1.21. Brakes

Braking requirements are based on UNECE Regulation 13-H.

- 1.21.1. The braking system must be approved by the team's certifying engineer.
- 1.21.2. The Competition Vehicle must be equipped with a service braking system that enables the driver to control the Competition Vehicle and bring it to a halt without removing either hand from the steering control.
- 1.21.3. The Competition Vehicle must be equipped with a secondary braking system, operated by the same control as the service braking system, that the driver can use to control the Competition Vehicle to a halt if the service braking system suffers any single failure.

The service braking system in a conventional vehicle is controlled by a brake pedal that operates two independent hydraulic brake circuits. One possible configuration has the front wheels on one circuit and the rear wheels on the other. An alternative configuration has a circuit for each diagonal pair of wheels. These arrangements meet the requirements for service and secondary braking systems—the service system operates both hydraulic circuits, and the secondary system operates whichever circuit has not failed.

- 1.21.4. Service and secondary braking systems may share components deemed 'not liable to failure' if they are amply dimensioned and readily accessible for maintenance. Components 'not liable to failure' are:
 - A brake pedal and its bearing
 - Hydraulic cylinders and their pistons
 - Hydraulic control valves
 - Brake cylinders and their pistons
 - Brake lever and cam assemblies
- 1.21.5. Hydraulic brake hoses and lines are regarded as liable to failure.
- 1.21.6. The service braking system must apply braking effort to all road wheels.
- 1.21.7. Braking must not cause the Competition Vehicle to yaw. This requirement applies to both the service braking system and the secondary braking system.
- 1.21.8. For Competition Vehicles without anti-lock brakes, the front wheels must lock up before the rear wheels.
- 1.21.9. The service braking system must be able to stop the fully laden Competition Vehicle within a distance

$$0.1 v + 0.0060 v^2$$





meters from any speed v , in km/h, that the Competition Vehicle can achieve.

If regenerative braking is disabled when the energy storage system is full then the service braking test must be done with no regenerative braking.

- 1.21.10. If the service braking system fails, the secondary braking system must be able to stop the fully laden Competition Vehicle within a distance

$$0.1 v + 0.0158 v^2$$

meters from any speed v , in km/h, that the Competition Vehicle can achieve.

Braking the single rear wheel of trike, for example, might not provide sufficient braking force.

- 1.21.11. Competition Vehicles must be equipped with a parking brake that can be operated by the driver from the normal driving position. The parking brake must hold the fully laden Competition Vehicle on a 20% incline or decline.

1.22. Tyres

- 1.22.1. Tyres must be suitable for highway use and used in accordance with their manufacturer's recommendations always.

- 1.22.2. Competition Vehicles must be fitted with tyres that are:

- Compliant with UNECE Regulations 30, UNECE Regulation 75 or US FMVSS 571.109, as indicated by an E or DOT approval marking on the tyre; or
- Otherwise approved by the event organisers

Experimental or prototype tyres must be approved by the South African road traffic authorities. The event organisers will administer this process, which requires the tyre manufacturer to submit a sample tyre and written technical specifications to the event organisers' office not later than 27 March 2026. A positive outcome to the approval process, which may take up to three months, is not guaranteed.

- 1.22.3. The speed rating of the tyres must be more than the maximum speed of the Competition Vehicle. The load rating of each tyre must be more than the maximum static load imposed on it by the fully laden Competition Vehicle.

- 1.22.4. Tyres must be approved by the team's certifying engineer.

- 1.22.5. Tyres must be free of any apparent defect.

1.23. Driving and Reversing

- 1.23.1. The fully laden Competition Vehicle must be able to start from rest on a 7% gradient.

- 1.23.2. The Competition Vehicle must be able to be driven backwards under its own power with the driver seated in the normal position.

1.24. Lighting

- 1.24.1. All Competition Vehicles must be fitted with:





- Two rear stop lamps
- One central stop lamp
- Left and right front direction indicator lamps
- Left and right-side direction indicator lamps
- Left and right rear direction indicator lamps
- Two daytime running lamps
- Two headlamps

During the 2024 Sasol Solar Challenge, foul weather was observed on more than one occasion, especially during early mornings and late afternoons. Competition Vehicles fitted only with daytime running lamps were far less visible to other road users and had visibility restrictions while driving. As a necessary precaution for the safety of all participants and Competition Vehicle drivers and passengers, it is a requirement for each Competition Vehicle to be fitted with headlamps. This will increase visibility for other road users as well as for Competition Vehicle drivers when driving in fog, mist, rain, low light and overcast conditions on South African public roads.

- 1.24.2. Stop lamps must emit red light. Direction indicator lamps must emit amber light. Daytime running lamps and headlamps must emit white light.

If you are designing your own lamps, then you must show that your lamp colours conform to the chromaticity requirements described in UNECE Regulation 48.

- 1.24.3. Lamps must be compliant with UNECE Regulations 6, 7, and 87, or the SAE/DOT equivalents. Teams must demonstrate compliance by either:
- The presence of compliance markings on the lamps, or
 - Detailed documentation that demonstrates compliance with the photometric requirements of the UNECE or SAE/DOT regulations, confirmed by the team's certifying engineer.

Lamps approved for motorcycles may not meet these requirements.

Photometric requirements for lamps are listed in Appendix 3 of UNECE Regulation 148. You may demonstrate compliance by testing or by calculations. If you are using calculations, be aware that most LEDs do not emit the same intensity at every angle, so the simple formula for converting luminous flux (lumens) to luminous intensity (candelas) does not apply.

- 1.24.4. Competition Vehicles must have the correct type of lamp in each position. Lamps must be mounted with the correct orientation so that the photometric requirements of UNECE Regulations 6, 7, and 87, or the SAE/DOT equivalents, are met.



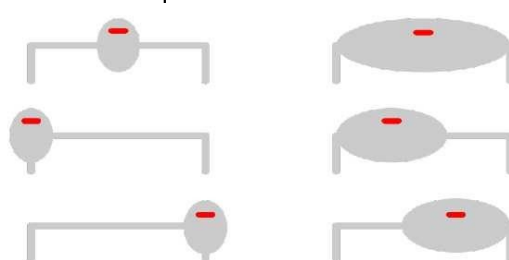
Lamp	UNECE Category	SAE/DOT Type
Front Indicators	1, 1a, 1b	I3, I4, I5
Rear Indicators	2a, 2b	I6
Side Indicators	5, 6	E2
Stop Lamps	S1, S2	S
Central Stop Lamp	S3	U3
Daytime Running Lamp	RL	Y2
Headlamps	HL	J

Lamp position and visibility requirements are based on UNECE Regulation 48. 'Apparent surface' is defined in the glossary and in UNECE Regulation 48.

1.24.5. The rear stop lamps must meet the following position and visibility requirements:

- for each of the left and right stop lamps, the lateral distance between the outer edge of the apparent lamp surface and the corresponding outer edge of the Competition Vehicle must be not more than 400 mm
- the minimum lateral distance between the apparent surfaces of the two stop lamps must be at least 600 mm, or at least 400 mm if the Competition Vehicle is less than 1300 mm wide
- the lowest point on the apparent surface of each stop lamp must be at least 350 mm above the ground
- the entire apparent surface of each lamp must be visible from every point in the region between the planes 15° up, 5° down, and 45° to the left and right.

1.24.6. A central stop lamp is required. Viewed from behind the Competition Vehicle, the lateral position of the lamp must coincide with the visual centre of the Competition Vehicle (see the examples in the following diagram). The lowest point on the apparent surface must be higher than a point 150 mm below the rear windscreen if the Competition Vehicle has a rear windscreen, and the





bottom of the apparent surface of the lamp must be higher than the top of the apparent surface of the rear stop lamps. The entire apparent surface must be visible from every point in the region between the planes 10° up, 5° down and 10° to the left and right.

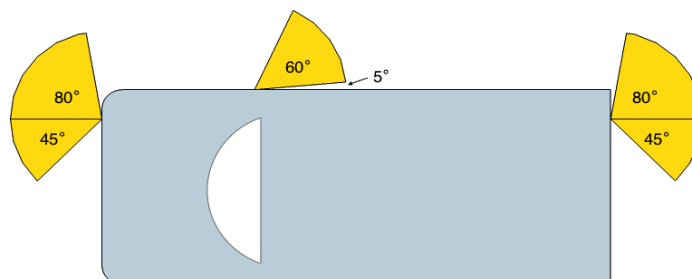
1.24.7. The following position requirements apply to the left and right front indicator lamps and to the left and right rear indicator lamps:

- for each of the left and right indicator lamps, the lateral distance between the outer edge of the apparent lamp surface and the corresponding outer edge of the vehicle must be not more than 400 mm
- the minimum lateral distance between the inner edges of the apparent surfaces of the left and right indicator lamps must be at least 600 mm, or at least 400 mm if the Competition Vehicle is less than 1300 mm wide
- Side indication indicator lamps must be less than 1800 mm behind the front most part of the Competition Vehicle and within 400 mm of the extreme outer edge of the Competition Vehicle on each side.
- the lowest point on the apparent surface of each indicator lamp must be at least 350 mm above the ground.

1.24.8. Side direction indicator lamps must be less than 1800 mm behind the front most part of the Competition Vehicle, and the outer edge of each apparent surface must be within 400 mm of the extreme outer edge of the Competition Vehicle on each side.

1.24.9. The entire apparent surface of each front and rear direction indicator lamp must be visible from every point in the region between the planes 15° up, 5° down, 45° inboard and 80° outboard (see the diagram below, which shows the horizontal angles for the right indicators).

1.24.10. The entire apparent surface of each side indicator must be visible from every point in the region between the planes 15° up, 5° down, 5° outboard and 65° outboard (see the diagram below).



1.24.11. Direction indicators must flash at 90±30 flashes per minute.



- 1.24.12. It must be possible to flash left and right direction indicator lamps simultaneously, as a hazard warning signal.
- 1.24.13. Daytime running lamps must be mounted at the front of the Competition Vehicle so that:
- the minimum lateral distance between the apparent surfaces of the lamps is at least 600 mm, or at least 400 mm if the Competition Vehicle is less than 1300 mm wide
 - the lowest point on each apparent surface is at least 250 mm above the ground
 - the entire apparent surface is visible from every point in the region between the planes 10° up, 10° down, 20° outwards and 20° inwards.
- 1.24.14. Headlamps must be at the front of the Competition Vehicle so that
- the lateral distance between the outer edge of the apparent surface and the corresponding outer edge of the Competition Vehicle is not more than 400 mm
 - the minimum distance between the apparent surfaces is at least 600 mm, or at least 400 mm apart if the Competition Vehicle is less than 1300 mm wide
 - the lowest point on each apparent surface is at least 500 mm above the ground
 - each entire apparent surface is visible from every point in the region between the
 - planes 15° up, 10° down, 45° outwards and 10° inwards.
- 1.24.15. The stop lamps must operate whenever driving is possible, and the brakes are applied.
- 1.24.16. The daytime running lamps must operate whenever driving is possible.
- 1.24.17. The headlamps must operate whenever driving in low visibility conditions or foul weather. Ideally, headlamps must be operated whenever driving is possible.
- 1.25. Audible Warning Device**
- 1.25.1. An audible warning device complying with the intent of UNECE Regulation 28 must be fitted to the Competition Vehicle.
- 1.25.2. Sound pressure level must be more than LA = 105 dB measured 2 m from the horn.

The horn should be mounted so that Competition Vehicle occupants are not subjected to excessive sound pressure levels.





A standard car horn is unlikely to achieve 105 dB if it is sealed inside the Competition Vehicle and not open to the air outside the Competition Vehicle.

- 1.25.3. The device must emit a continuous and uniform sound. The audible warning device must be capable of operating for 50,000 cycles of on for one second and off for four seconds.

1.26. Instrumentation

- 1.26.1. The following information must always be visible to the driver (i.e., not behind menus) while the Competition Vehicle is in motion.
- The speed of the Competition Vehicle
 - Whether the direction indicators are operating
 - Whether the hazard lights are operating
 - Energy storage system warnings
 - Electronic rear vision images (if fitted)
- 1.26.2. This instrumentation must be powered from the energy storage system, and not from separate batteries.

1.27. Automatic Functions

- 1.27.1. Any cruise control function must automatically deactivate when the brake is operated, or the car is turned off.

Cruise control must not resume automatically following a brake operation.

- 1.27.2. Any automatic driving function must immediately deactivate on manual input or when the car is turned off.

Automatic driving must not resume automatically.

1.28. Electrical Safety

Electrical safety requirements are based on Section 5 of UNECE Regulation 100. The term “high voltage” means more than 60 V dc or more than 30 V rms ac.

- 1.28.1. Protection against direct contact with high-voltage parts, including conductors, must be achieved using double insulation, enclosures, or barriers. It must not be possible to remove protection without the use of tools.
- 1.28.2. Protection against direct contact with high-voltage parts inside the driver, passenger and luggage compartments must be designed to exclude objects larger than 1 mm diameter (Ingress Protection rating IPXXD).
- 1.28.3. Protection against direct contact with high-voltage parts inside areas other than the driver, passenger, and luggage compartments must be designed to exclude fingers (Ingress Protection rating IPXXB).





- 1.28.4. Double insulation must meet the AS 3001/IEEE 100 definition: comprising both basic insulation and independent supplementary insulation that provides protection equivalent to that of the basic insulation. A single layer of reinforced insulation is also acceptable if it provides protection equivalent to double insulation.

Electrical tape and flimsy, easily displaced covers are unlikely to meet the requirements of supplementary insulation. The front surface of a photovoltaic cell is deemed to be double-insulated if properly encapsulated. The rear surface is deemed to be 'live' unless the team can demonstrate otherwise.

- 1.28.5. High-voltage energy storage packs must be marked with the high-voltage symbol shown in the following diagram.



- 1.28.6. The high-voltage symbol must also be visible on any enclosure or barrier that can be accessed without using tools if removing the enclosure or barrier exposes high-voltage parts.

Example: A motor controller contains high voltage parts, so their parts must be protected by an enclosure or barrier that requires tools to remove (Technical Regulation 1.28.1). if it is possible to access the motor controller enclosure without using tools (e.g., by opening the boot or tilting the solar collector) then the motor controller enclosure must have a high-voltage symbol on it.

- 1.28.7. The resistance between any exposed conductive part and each terminal of the energy storage system must exceed 100 V ohms, where V is the nominal voltage of the energy storage system.

This is equivalent to a maximum leakage current of 10 mA.

- 1.28.8. The resistance between any exposed conductive part and each terminal of every solar cell must exceed 100 V ohms, where V is the maximum circuit voltage of the solar collector.
- 1.28.9. A system must be implemented to monitor the isolation of the Competition Vehicle's chassis and body as per Technical Regulations 1.28.7 and 1.28.8. this system may take the form of a permanently connected electronic device or a properly documented procedure of regular manual checks.
- 1.28.10. A fuse or circuit breaker, suitably rated to protect the wiring, must be mounted in or on each energy storage pack. Additionally, the dc interrupting current



(i.e., dc breaking current) capacity of the device must be able to interrupt the maximum possible short circuit current of the battery packs.

A typical Competition Vehicle might use a 50 A fuse with a 15000 A dc interrupt rating. Fuses and circuit breakers intended for ac use might not meet this requirement. Teams must provide a manufacturer's datasheet clearly showing that the dc specifications of the device are adequate.

1.29. Electrical Safe State

The Regulations in this section are for a Competition Vehicle with a photovoltaic solar collector and electrochemical energy storage. Teams using alternative solar collector or energy storage technologies must ensure that their Competition Vehicle can be made safe in an emergency. The systems used to achieve this must meet equivalent levels of safety based on the organiser's assessment of the proposal submitted by the entrant.

1.29.1. The Competition Vehicle must have a "safe state" which, in an emergency, minimises the risk of electrical fire and electrical shock to occupants, team members, emergency response personnel, and bystanders. When in safe state:

- Every conductor emerging from each energy storage pack must be galvanically isolated from every energy storage cell. This means that there must be a contactor in series with the negative pole of the battery and a second contactor in series with the positive pole, in each energy storage pack.
- No voltage may be present across any pair of conductors emerging from any energy storage pack.
- No voltage may be present across any pair of conductors emerging from any solar collector. This isolation must be performed with a contactor, which must be mounted on the same structural component as the solar collector it is associated with.
- All contactors described above must immediately and automatically open when safe state is invoked.
- No current may be present through any conductor loop that is external to the energy storage packs or the solar collectors.

MOSFETS and other semiconductor devices are not considered to offer galvanic isolation.

Safe state is for emergencies and for complete shutdown of the Competition Vehicle. In addition to safe state, a Competition Vehicle may have a 'standby' state that provides power to some subsystems outside of the energy storage packs.





- 1.29.2. Any part of any conductor that is more than 200 mm from the nearest PV cell is considered to be outside of the solar collector.

This Regulation applies regardless of the solar collector voltage. Emergency first responders are told that any wiring within 200 mm of a PV cell may be live when the Competition Vehicle is in safe state, and no wiring more than 200 mm from a PV cell will be live when the Competition Vehicle is in safe state.

- 1.29.3. All mechanisms for placing the Competition Vehicle into safe state and maintaining safe state must be fail-safe; if an electrical activation mechanism fails, the Competition Vehicle must automatically and immediately place itself into safe state and must remain in safe state until there is a deliberate intervention by the Team.
- 1.29.4. The safe state circuit must be designed to be as simple and direct as possible. In particular, it must be designed in such a way that failure of a semiconductor device or devices, no matter how improbable, cannot:
- prevent the Competition Vehicle from entering safe state when the activation device is operated, or
 - bring the Competition Vehicle out of safe state without deliberate driver intervention.
- 1.29.5. The driver must be able to place the Competition Vehicle into safe state with a single simple action while seated in the normal driving position and without releasing the safety belt. This activation device must be clearly labelled.
- 1.29.6. For emergency use, an activation device that operates with a single simple action that immediately places the Competition Vehicle into safe state must be provided on the exterior of the Competition Vehicle. Multiple activations must not bring the Competition Vehicle out of safe state. The activation device must be placed within a yellow disc with a diameter of 180 mm. Also, in the yellow disk must be a blue equilateral triangle (minimum side length 150 mm) that contains a red flash, with the legend Emergency Electrical Isolation. In addition, there must be a clear instruction on how to operate the device (e.g., PULL or PRESS). The yellow isolation disc and the activation mechanism must be clearly visible to an emergency services first responder approaching the driver and must be within 100 mm of the lower edge of the windscreen,





adjacent to the driver egress opening. No part of the yellow disc may be behind the driver.



Both activation mechanisms must be demonstrated at static scrutineering. The scrutineers may invite random members of the public to validate the ease of operation of the external emergency activation device.



Glossary

Battery	Electrochemical cells wired in series or parallel and housed in a single container
BWSC	Bridgestone World Solar Challenge
Certifying Engineer	A professional engineer engaged by a team to report on compliance with regulations and roadworthiness requirements. The certifying engineer must be qualified to certify vehicles for operation on public roads.
Clerk of the Course	The person responsible for coordinating Dynamic scrutineering and the on-road segment of the event
DST	South African Department of Science and Technology
Energy Storage System	The Competition Vehicle subsystem used to store energy, comprising one or two energy storage packs and the electrical connections between them.
Event Name	The official name of the event is the "Sasol Solar Challenge"
FIA	Federation Internationale de'Automotive
High Voltage	More than 60 V dc or more than 30 V rms ac
Judge	A person invited to make subjective comment on Cruise Class attributes
Juror	A person appointed by the event to resolve disputes
Participant	A person who has registered to participate in the event as a member of a team
Regulations	2024 Sasol Solar Challenge Sporting Regulations, Technical Regulations and Team Manager's Guide
Official Website	http://www.solarchallenge.org.za
Official Facebook	http://www.facebook.com/SASolarChallenge
Official Time	South African Standard Time GTM +2





Solar Collector

The Competition Vehicle subsystem used to collect solar energy. It is typically an array of photovoltaic cells, with or without concentrators or reflectors, but other types of solar collectors are possible

Steward

An event official responsible for ensuring regulations are applied correctly and fairly

Team Manager

The person in charge of, and responsible for, the actions of a team

UNECE

United Nations Economic Commission for Europe, responsible for regulations for motor vehicles